

ULAANGOM, Mongolia

WMO: 442120

Lat: 49.9714N

Lon: 92.0778E

Elev: 939

StdP: 90.54

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.3	-39.3	-45.1	0.0	-41.2	-42.8	0.1	-39.2	5.3	-31.3	4.7	-32.5	1.5	180	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	30.3	16.1	28.5	15.5	27.0	15.0	17.7	26.4	16.8	25.3	16.1	24.3	3.2	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.7	11.7	20.8	13.7	11.0	20.4	12.7	10.3	20.1	54.0	26.2	51.0	25.6	48.6	24.3	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	6.8	5.5	-42.0	33.6	2.5	2.1	-43.8	35.1	-45.3	36.3	-46.7	37.5	-48.5	39.0		
(5)			-42.0	19.2	2.5	1.3	-43.8	20.1	-45.3	20.9	-46.7	21.6	-48.5	22.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.8	-31.1	-27.9	-15.7	3.3	11.7	18.2	20.3	17.6	10.4	2.0	-8.8	-23.5	(6)
(7)		DBStd	18.60	3.89	5.07	6.77	5.97	4.51	3.59	2.84	3.23	4.06	4.19	5.55	6.23	(7)
(8)		HDD10.0	5271	1273	1061	797	209	35	1	0	0	43	249	566	1038	(8)
(9)		HDD18.3	7508	1531	1294	1055	451	209	46	13	52	238	507	816	1296	(9)
(10)		CDD10.0	955	0	0	0	8	88	246	319	237	56	1	0	0	(10)
(11)		CDD18.3	149	0	0	0	0	4	41	73	30	1	0	0	0	(11)
(12)		CDH23.3	1593	0	0	0	1	75	489	703	306	19	0	0	0	(12)
(13)		CDH26.7	404	0	0	0	0	11	124	197	70	2	0	0	0	(13)
(14)	Wind	WSAvg	2.2	1.1	1.3	1.7	2.6	3.2	2.9	2.5	2.5	2.5	2.3	1.9	1.3	(14)
(15)	Precipitation	PrecAvg	147	3	3	4	4	8	23	39	33	10	7	9	6	(15)
(16)		PrecMax	248	6	8	11	14	22	69	122	89	31	31	21	16	(16)
(17)		PrecMin	72	0	0	0	0	0	2	9	4	1	0	1	1	(17)
(18)		PrecStd	53	2	2	3	4	6	15	33	22	8	7	5	4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-19.3	-11.3	2.8	21.2	28.3	32.1	33.4	31.2	25.9	17.3	4.9	-7.9	(19)
(20)			MCWB	-19.9	-12.1	-0.2	9.3	14.2	16.1	18.1	16.3	14.1	8.0	1.4	-8.8	(20)
(21)		2%	DB	-21.1	-15.3	-0.7	18.2	25.5	29.8	30.8	28.6	22.9	13.9	2.3	-9.7	(21)
(22)			MCWB	-21.7	-15.9	-2.5	8.0	12.6	15.2	16.6	16.1	12.8	6.9	-0.3	-10.3	(22)
(23)		5%	DB	-22.8	-17.6	-3.0	15.6	23.1	27.9	28.9	26.6	20.3	11.6	0.3	-12.1	(23)
(24)			MCWB	-23.2	-18.2	-4.5	6.8	11.2	14.8	15.9	15.3	11.7	5.8	-1.6	-12.8	(24)
(25)		10%	DB	-24.5	-19.8	-5.2	13.1	20.8	26.3	27.2	24.7	18.4	9.5	-1.3	-14.2	(25)
(26)			MCWB	-24.8	-20.2	-6.2	5.5	10.1	14.2	15.6	14.9	10.6	4.5	-2.7	-14.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-19.8	-12.2	-0.1	10.0	15.0	17.7	19.9	18.4	14.7	8.9	1.6	-8.8	(27)
(28)			MCDWB	-19.5	-11.5	2.9	19.8	26.4	28.6	29.0	26.9	23.9	15.8	4.3	-8.3	(28)
(29)		2%	WB	-21.6	-15.8	-2.3	8.5	13.3	16.6	18.3	17.2	13.3	7.4	-0.1	-10.3	(29)
(30)			MCDWB	-21.2	-15.3	-0.8	17.4	24.2	26.4	26.8	25.4	21.7	13.1	1.8	-9.7	(30)
(31)		5%	WB	-23.2	-18.0	-4.3	7.1	11.9	15.7	17.3	16.4	12.1	6.1	-1.4	-12.8	(31)
(32)			MCDWB	-22.9	-17.5	-3.1	14.6	21.9	24.8	25.6	24.1	19.7	11.2	0.2	-12.2	(32)
(33)		10%	WB	-24.8	-20.2	-6.3	5.8	10.6	15.0	16.5	15.5	11.0	4.7	-2.7	-14.8	(33)
(34)			MCDWB	-24.6	-19.8	-5.1	12.6	19.7	24.1	24.7	23.0	17.7	9.2	-1.4	-14.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.1	11.0	11.0	10.7	13.0	12.7	11.4	11.6	12.4	11.2	7.2	7.4	(35)
(36)			MCDBR	9.9	12.1	8.8	14.9	16.1	15.2	14.1	14.1	14.4	13.8	8.9	6.1	(36)
(37)			MCWBR	9.5	11.5	7.2	7.1	7.2	5.5	4.8	5.3	6.9	8.3	6.7	5.5	(37)
(38)		5% WB	MCDBR	9.7	12.1	8.2	14.1	15.2	12.8	11.7	12.1	13.8	13.4	8.6	5.9	(38)
(39)			MCWBR	9.4	11.5	6.7	7.1	7.2	5.3	5.0	5.3	6.9	8.4	6.6	5.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.250	0.278	0.326	0.373	0.374	0.377	0.375	0.355	0.317	0.300	0.274	0.250	(40)	
(41)		taud	2.284	2.241	2.219	2.220	2.262	2.348	2.388	2.412	2.459	2.424	2.310	2.258	(41)	
(42)		Ebn at Noon	806	864	875	868	882	880	877	879	884	833	761	743	(42)	
(43)		Edh at Noon	67	92	114	128	128	119	113	105	89	77	66	60	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.35	2.35	3.87	5.41	6.53	6.73	6.38	5.62	4.46	2.81	1.41	1.03	(44)	
(45)		RadStd	0.08	0.14	0.19	0.23	0.34	0.26	0.36	0.28	0.21	0.16	0.12	0.08	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	-2.43	N/A	-0.61	-0.74	N/A	N/A	N/A	N/A	(46)
(47)	Regional (6 neighbors)	N/A	N/A	-2.56	N/A	-0.86	-1.16	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page